KryloCanvas (Web-WhiteBoard)

Created and Designed by Abhimat Chatterjee

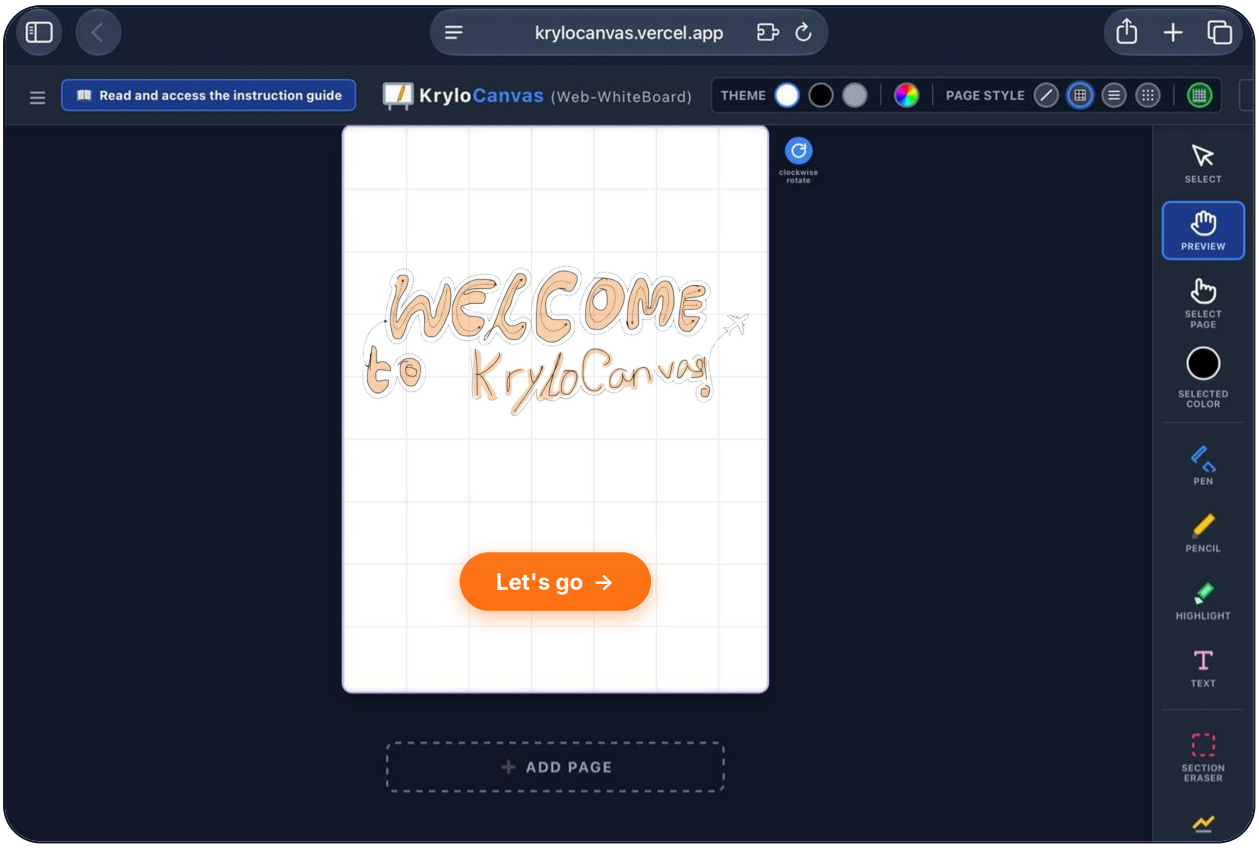


Image: KryloCanvas Start Screen

1. Executive Overview & Architecture

KryloCanvas (Web-WhiteBoard) is a professional, high-precision, browser-based digital whiteboard and mathematical drawing suite for classrooms, STEM coaching institutes and technical educators. It runs inside any modern browser with no installation, supports touch and stylus, and now opens very large PDF, Word, Excel and PowerPoint files without crashing thanks to its lazy-loading page engine.

Core Operational Strengths:

- **Zero Backend:** HTML5 Canvas, Fabric.js, PDF.js and jsPDF in one file.
- **Universal State Engine:** Colour and thickness sync across every tool.
- **Lazy Page Engine:** Huge files open fast, only a few pages awake at a time.

Mathematical & STEM Capabilities:

- **Solid Instruments:** Ruler, Protractor and Set Square with pen snapping.
- **Precision Compass:** Fixed needle, exact radius in cm, mm or inch.
- **43 Shapes & Flowcharts:** Vector shapes, process flows and trees.

QUICK START (6 STEPS)

1. Open **krylocanvas.vercel.app**; on the welcome screen tap **Let's go**.
2. Pick a theme and page style in the top bar; tap **Add Page** for more pages.
3. Choose Pen, Pencil, Highlighter or Text; set colour and thickness.
4. Use Rulers, Compass, Shapes or Flowchart for precise figures.
5. **Import** a PDF, image, Word, Excel or PowerPoint file to annotate it.
6. **Export as** PDF (all or current page) or image; optionally rename.

NEW IN THIS VERSION

- **Lazy loading:** byte-range PDF reading, max 5 live pages, page cache.
- **Word / Excel / PPT import** through the same lazy pipeline.
- **Custom Theme Colour** (spectrum, RGB, HEX); **Select Page** tool.
- **Page Rotate** (90° clockwise) and per-page **Delete**.
- **Rename Export** prefix; **Welcome** start screen.

2. Workspace Layout: Header, Pages & Tool Palette

KryloCanvas has three zones designed for minimum cognitive load while teaching: a **top command header**, a scrolling **pages area** (with a dashed **+ Add Page** button) and a **right-hand tool palette** (a horizontal strip at the bottom on phones).

Top Command Header

- **Instructions** (info icon): in-app help and the QR / link to this guide.
- **Theme dots:** White, Black, Grey and a Custom Theme Colour spectrum button.
- **Page style buttons:** Blank, Standard Grid, Single Lines, Dotted Grid, plus the Millimeter Graph button (5 colours).
- **Import** and **Export as** drop-down menus.
- **Undo / Redo**, **Refresh** and **Close** (both ask for confirmation).

Pages Area & Tool Palette

- Pages are stacked vertically (1200 × 1600 px each) and scroll freely.
- Every page has its own **Rotate** (clockwise) and **Delete** buttons.
- Palette: Select, Preview, Select Page, colour indicator, Pen, Pencil, Highlighter, Text, two Erasers, Shapes, Rulers, Compass, Flowchart, Clear All and the thickness slider.
- Welcome screen on launch: tap **Let's go** to enter the board.

3. Command Bar & Header Reference

Control	Where	What it does
Instructions	Header, left	Opens the help panel and the button / QR to open or download this PDF guide.
White / Black / Grey	Header theme dots	Switches page background; strokes, grid and page stamp invert automatically.
Custom Theme Colour	Header, spectrum dot	Opens colour picker (spectrum, R G B, HEX, saturation, value, contrast, quick shades); changes apply live.
Blank / Grid / Lines / Dots	Header page styles	Changes the paper pattern of the active page (grid, ruled lines or dotted matrix).
Millimeter Graph	Header, graph dot	Adds a new graph-paper page; choose Green, Blue, Red, Orange or B&W.
Undo / Redo	Header	30 steps per page; works on the page you last touched.
Refresh / Close	Header, right	Reload or close the session only after a confirmation dialog.
+ Add Page	End of pages	Appends a new blank page in the chosen theme and updates all page counters.
Rotate / Delete	On each page	Turns that page 90° clockwise with its drawings; or removes the page.

4. Import & Export Menus

Import (file menu)

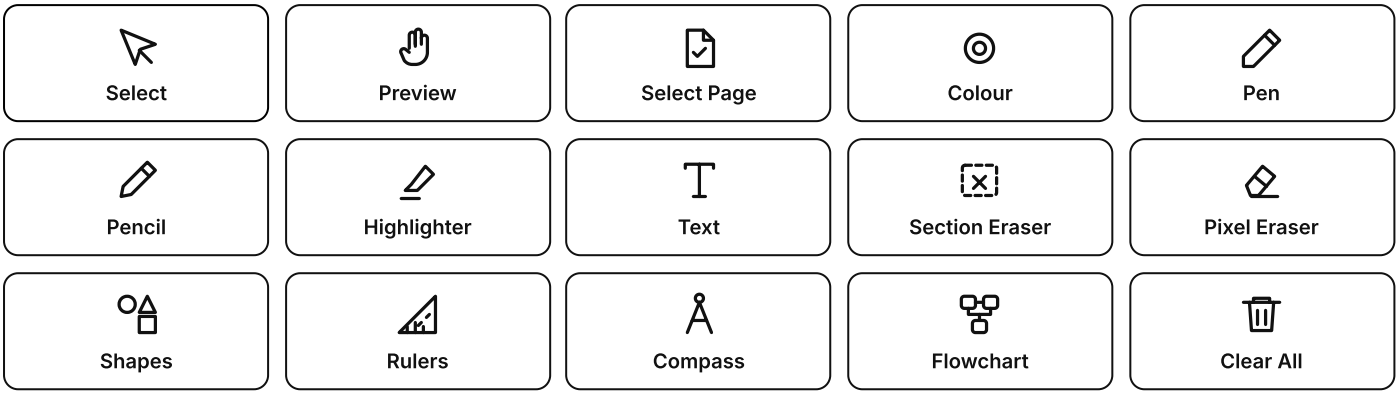
- **PDF** – every page becomes an editable whiteboard page (lazy loaded).
- **Image** – PNG / JPG placed on the page; move, scale, rotate, delete.
- **Word** – .docx (MS Word or Google Docs download), auto-paginated.
- **Excel** – .xlsx, .xls, .csv; 34 rows × 7 columns per page.
- **PowerPoint** – .pptx slides rendered as pages.

Export as

- **PDF (Entire Document)** – all pages, one at a time, to protect memory.
- **PDF (Current Page)** – only the active page.
- **Image (Current Page)** – PNG of the active page; rulers and compass are hidden.
- **Rename Export** dialog: type an optional name; spaces become underscores. Press **Export**, or **Default name** to skip.

5. The Tool Palette at a Glance

The palette on the right lists every tool in a fixed order. Tap a tool once to activate it; the active tool is highlighted with a frame. The universal **thickness slider** (1–40) sits at the bottom.



6. Navigation: Select, Preview & Select Page

Select Tool

- Tap any object to get its bounding box.
- Move, scale, rotate (top handle) or delete drawings, text, shapes, images and flowchart blocks.
- Drag on empty space to marquee-select a group.
- Instruments and images show a **Delete** (top-right) and **Rotate** (top-centre) button.

Preview Tool (Hand)

- Locks drawing so touches only move the view.
- Pinch in / out, or **Ctrl + scroll** on a computer, to zoom from 50 % to 400 %.
- Drag to pan across the pages.
- Pinching on the top or side menus zooms only the menus.

Select Page Tool

- Tap a page to make it the **active page**.
- Drawing is disabled while it is active, so nothing is inked by accident.
- Active page decides where shapes, instruments and Undo / Redo apply.
- Use it before inserting a figure onto a particular page.

7. Selected Colour Popup & Thickness

The round **colour indicator** in the palette shows the live stroke colour. Tapping it opens the Selected Colour popup:

Quick Swatches & Custom Colour

- **10 quick swatches:** Black, Red, Blue, Green, Yellow, Orange, Purple, Pink, Cyan, Grey.
- **Custom colour:** colour wheel / spectrum, saturation and value (HSV), RGB values and a HEX field.
- **Eyedropper:** pick any colour from the screen.
- **Recent colours** are remembered for fast reuse.

Universal Thickness (1–40)

- One slider drives the Pen, Pencil, Highlighter, Shapes and Compass output.
- **Pen:** width = thickness. **Pencil:** half of thickness (minimum 1).
- **Highlighter:** 4 × thickness at 40 % opacity so writing stays visible underneath.
- Change colour or thickness at any time; selected objects can be restyled with Select.

Why the engine is called “Universal”

One shared state holds the colour, thickness and active tool. Switching from Pen to Shapes or Compass never resets your style, so a teacher can keep a chosen colour for a whole lesson, and a single change updates what the next tool draws.

8. Pen, Pencil & Highlighter

Pen (Natural Ink)

- Continuous bold vector strokes in the selected colour.
- Best for stylus writing, derivations and headings.
- `width = thickness`

Pencil (Technical)

- Fine graphite-style line.
- Ideal for construction lines, geometry sketches and diagrams.
- `width = max(1, thickness / 2)`

Highlighter

- Translucent wash that never hides what is below.
- Works over text, formulas and imported pages.
- `width = 4 × thickness, opacity 40 %`

Drawing near a ruler, protractor, set square or compass with any of these three tools **snaps** the stroke to the instrument edge (see Section 16). Every stroke is stored as a vector object, so it can be selected, moved, resized and exported without blur.

9. Eraser Suite: Section vs Pixel Eraser

Eraser	How you use it	Behaviour	Best for
Section Eraser	Drag a dashed rectangular box across the page.	On release, every object enclosed by the box is deleted in one step (one Undo restores it).	Clearing whole proofs, diagrams or notes.
Pixel (Scribble) Eraser	Scribble over the line to be removed.	Round brush erases the part of the stroke it touches and leaves the rest.	Fixing a minus sign, an exponent or one small line.

10. Text Tool

Writing Text

- Choose **Text**, tap anywhere on a page and type plain text.
- It takes the current colour; **Enter** starts a new line.
- Tap existing text with the Text tool to edit it.
- An empty text box is removed automatically when you tap away.

Moving & Resizing Text

- Switch to **Select** to move, scale or rotate text.
- Text stays vector, so it prints crisply in the exported PDF.
- Delete selected text from the Select tool controls.

11. Clear All, Undo & Redo

Clear All Drawings

- The red trash button at the bottom of the palette wipes the drawings of the current page.
- Page background, theme and header stamp stay; use Undo straight after if pressed by mistake.

30-Step History per Page

- Each page owns its own Undo / Redo stack of 30 steps.
- The header buttons act on the page you last drew on.
- Importing or deleting pages does not disturb other pages' history.

Pro tips for neat handwriting

- Use **Preview** to zoom to 200 % before writing small formulas.
- Rest your palm freely in Preview or Select Page; no stray ink.
- Write with Pencil, then highlight key lines with the Highlighter.
- Use Section Eraser for big clean-ups and Pixel Eraser for fine edits.

12. Indelible Header Stamp: Date, Day & Page Counter

Every page carries a two-line stamp at the top-right corner so printed exports stay in order:

- **Line 1:** date and weekday, e.g. `07/11/2026 - Saturday`, read from the device clock.
- **Line 2:** page number over total pages, e.g. `4 / 10`.
- Counters renumber automatically when pages are added or deleted.
- Stamp colour inverts with the theme: dark on light pages and light on dark pages.

07/11/2026 - Saturday
4 / 10

13. Themes & Custom Theme Colour

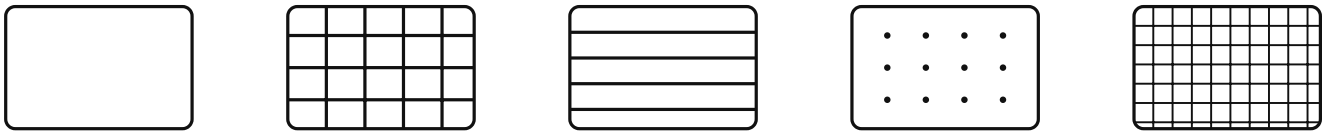
White, Black & Grey

- **White:** clean page for print-ready work.
- **Black:** dark page for projectors; black strokes flip to white automatically.
- **Grey:** neutral studio background.
- Switching theme recolours grid, lines, dots and the stamp.

Custom Theme Colour

- Open the spectrum dot to pick **any** page colour.
- Controls: colour spectrum, white-to-black brightness, R / G / B, HEX, saturation %, value %, contrast and quick shades.
- Applies live while picking.
- Grid, lines and dots become a light tint on dark pages and a dark shade on light pages.

14. Page Styles & 5 Millimeter Graph Papers



Blank | Standard Grid | Single Lines | Dotted Grid | Millimeter

Pick a page style in the header to restyle the active page. The **Millimeter Graph** button adds a *new* page of engineering graph paper in one of five colours: **Green, Blue, Red, Orange** or **Black & White** (high-contrast, ideal for laser printing). Colours invert automatically with the theme.

15. Multi-Page Sessions: Add, Rotate, Delete & Safety

Add & Number

- **+ Add Page** appends a page; Millimeter Graph appends graph pages.
- Pages are 1200 × 1600 px, portrait.
- The stamp “n / total” updates on every page.

Rotate & Delete

- The round arrow on a page turns it **90° clockwise**; its drawings turn with it. Press again for 180°, 270°.
- The bin button removes just that page.

Safety Dialogs

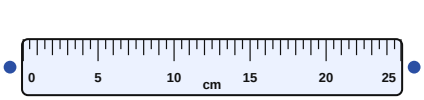
- **Refresh** and **Close** ask before acting.
- Export never deletes your work.
- Undo history is per page and capped at 30 steps.

✓ Recommended classroom flow

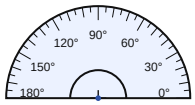
Choose theme and paper → add pages for each topic → write with Pen / Pencil → add figures with Shapes, Rulers and Compass → export **PDF (Entire Document)** and share. The date stamp and “n / total” make every handout traceable.

16. Solid Instruments & Snapping Engine

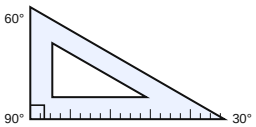
Open **Rulers** in the palette and choose **Ruler**, **Protractor** or **Set Square**. Each instrument appears once per page as a solid, movable object.



Ruler: stretch from the end handles



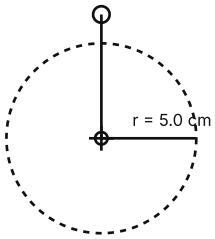
Protractor 0°–180°



Set Square

Topic	How it works
Solid edges	Instruments act as real boundaries: trace with Pen, Pencil or Highlighter and the stroke snaps to the edge, so lines and arcs are perfectly straight or curved.
Snapping range	The pen snaps when it is within 100 px of an instrument edge; keep your stylus near the edge.
Scale	1 cm = 37.8 px and 1 inch = 96 px, matching the on-screen ruler marks.
Ruler	Resize with the left / right handles; its marks regenerate crisply at the new length.
Protractor & Set Square	Move, rotate and delete with the on-object controls; trace for angles, perpendiculars and right-angle triangles.
Controls	Select an instrument: red **Delete** at the top-right, blue **Rotate** at the top-centre. Instruments are hidden in image exports.

17. Precision Compass System



- **Fixed needle:** the red crosshair marks the pivot; drag the compass with **Select** to place it anywhere on any page.
- **Radius:** 0.5 to 500 cm in 0.5 cm steps, with a numeric box for an exact value and a cm / mm / inch readout.
- **Circle:** draws a full 360° circle at the current radius.
- **Arc (90°):** draws an exact 90° arc along the compass arm; rotate the arm 15° at a time.
- **Snapping:** with Pen, Pencil or Highlighter near the compass, strokes lock on to the exact circumference.

HUD control	Action
− 0.5 cm / + 0.5 cm	Shrinks or enlarges the radius by half a centimetre.
Exact radius box	Type any value from 0.5 to 500 and choose the unit.
⟲ 15° / ⟳ 15°	Rotates the compass arm 15° counter-clockwise or clockwise.
Circle / Arc (90°)	Stamps a circle or arc using the universal colour and thickness.
Close (×)	Removes the compass from the page.

18. Classical Constructions with the Compass & Ruler

Because the radius is exact and the pen snaps to the arc, standard constructions can be drawn quickly. Use **Pencil** for construction arcs and **Pen** for the final figure.

Bisect a Line Segment

- Draw segment AB with the Ruler.
- Set a radius larger than half of AB.
- Place the needle on A and draw an arc; repeat from B.
- Join the two crossing points: this is the perpendicular bisector.

Equilateral Triangle

- Draw base AB with the Ruler.
- Set the radius equal to AB.
- Draw an arc from A and another from B.
- Join the crossing point C to A and B.

Bisect an Angle

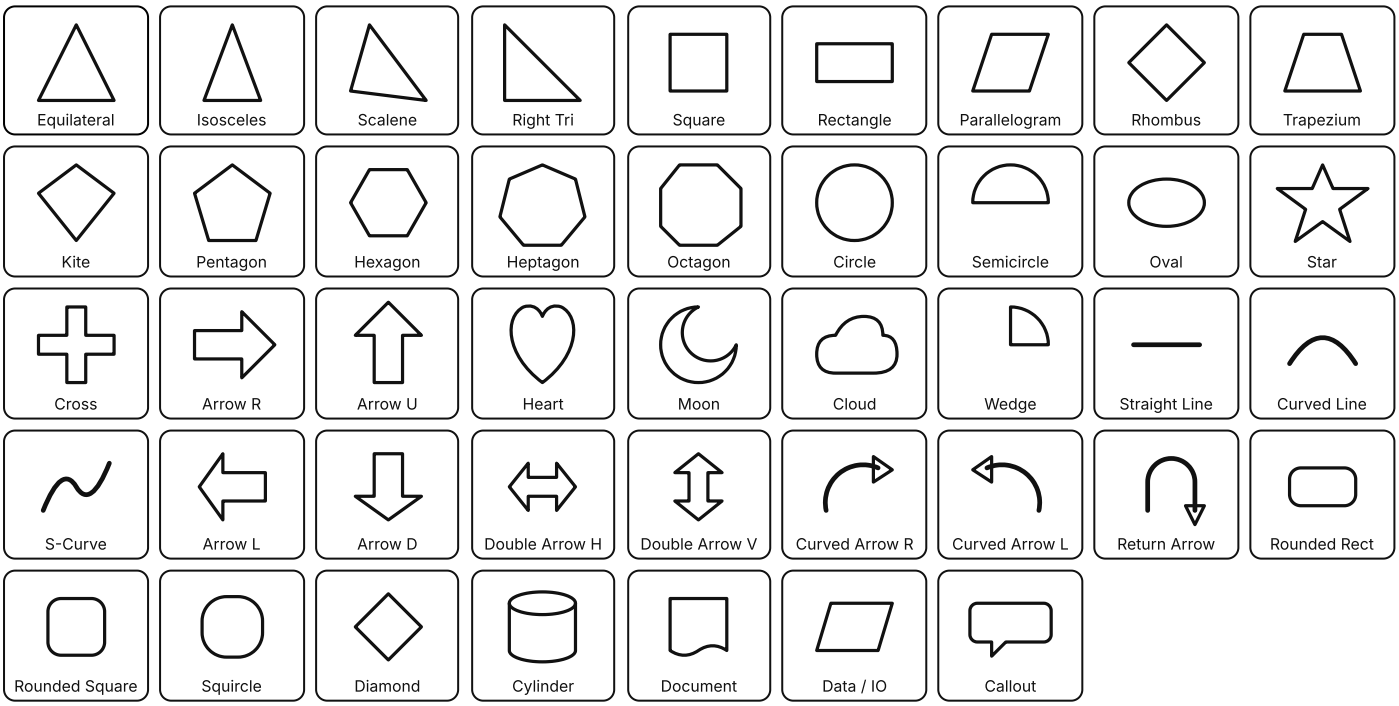
- Draw an arc from the vertex cutting both arms.
- From each cut point draw an arc of equal radius.
- Join the vertex to where the arcs cross.

Regular Hexagon in a Circle

- Draw a Circle of radius r.
- Without changing r, step round the circle with Arc cuts.
- Join six points with the Ruler for a hexagon.

19. The Extended 43-Shape Library

Tap **Shapes** to open the library, then tap a shape to drop it on the active page in the current colour. Shapes use the same universal thickness as ink and remain fully editable.



Groups

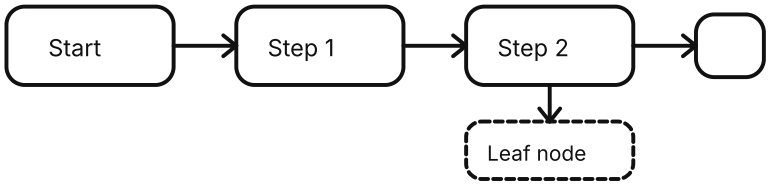
- **Polygons:** triangles, quadrilaterals, pentagon to octagon.
- **Curves:** circle, semicircle, oval, moon, cloud, heart, wedge.
- **Arrows & lines:** straight, curved, S-curve, single / double / curved / return arrows.
- **Flow symbols:** rounded rectangle, diamond, cylinder, document, data / IO, callout.

Edit After Inserting

- Select to move, scale and rotate with the corner handles.
- Change colour or thickness and press the shape again to restyle new shapes.
- Shapes snap like ink when drawn along instruments.
- Delete with the on-object control or the Section Eraser.

20. Flowchart & Process Tree Builder

Open **Flowchart** in the palette, set up the diagram in the dialog and press **Insert Flowchart on Page**. Blocks, arrows and text all use crisp vector shapes.



Dialog option	Choices
1. Direction	Horizontal Process Flow (left → right) or Vertical Tree (top → down).
2. Structure	**Process Flow** (sequential steps), **Tree Hierarchy** (root and leaves) or **Single Node** (extend manually).
3. Steps / Columns	Use – and + to set 1 or more steps (default 3).
4. Block Theme & Colour	Blue, Green, Purple, Orange, Slate or Custom colour for the blocks.

21. Editing in the Page: Floating Flowchart HUD

Flowchart HUD Buttons

- **Add Step** – adds a connected next block.
- **Leaf Node** – adds a subordinate branch under the selected block.
- **Text** – opens "Edit Flowchart Block Text".
- **Delete** – removes the selected block.
- **↔ / ↕ Dir** – toggles horizontal / vertical direction.
- **×** – closes the toolbar.

On-Block Controls & Gestures

- Selecting a block shows round handles: Add Connected Next Node, Add Subordinate / Leaf Node, Edit Block Text, Delete Block and Colour Theme.
- **Double-click** a block to rename it quickly.
- **Pull / push** the connector handles to grow or fold branches.
- Each block can have its own colour theme; a status line shows the active block.

22. Examples & Best Practices

Use cases

- Algorithm flowcharts and decision trees.
- Chapter mind-maps and family trees.
- Chemistry and biology process chains.
- Project stages and exam revision maps.

Best practices

- Keep block text short (2–4 words).
- Use Process Flow for steps and Tree for hierarchy.
- Pick one colour theme per diagram for printing.
- Use Select to move whole blocks; Undo reverses edits.

Combine tools

Flowchart blocks sit on the same page as ink, shapes and instruments. Annotate with the Pencil, highlight key blocks with the Highlighter and export the page to PDF, all without leaving the board.

23. Lazy-Loading Engine: Open Huge Files Without Crashing

Importing a large book, slide deck or workbook no longer loads everything into memory. KryloCanvas reads only what it needs, keeps just a few pages “awake”, and puts the rest to sleep. This keeps tablets and low-memory devices fast and stable.



How the PDF Engine Works

- **Byte-range loading:** only the PDF’s structure and the pages you need are read, not the whole file.
- **Page shells:** the first 20 pages appear at once; remaining shells are created in background batches.
- **≤ 5 awake pages:** about 3 visible plus one buffer on each side are rendered; others sleep.
- **Sleeping pages** shrink their canvas to 1 × 1 px to free memory, then restore when scrolled back.
- **Page cache:** up to 40 compressed page images (30 on low-memory devices) are kept for quick return.
- **Pixel cap:** each rendered page is limited to about 3.4 MP (2.3 MP on low-memory devices).

Safe Export of Big Documents

- PDF export loads and writes pages **one at a time**, then releases each page.
- Your drawings are kept on top of the original page, in vector quality.
- Choose **PDF (Current Page)** for a quick single sheet.
- The Rename Export dialog lets you add a name in front of the file name.

24. Word, Excel & PowerPoint Import

File type	How it is converted	Notes
Word (.docx)	Read by the built-in converter and split into 1200 × 1600 pages.	Works with MS Word and Google Docs downloads. Large embedded pictures are slimmed.
Excel (.xlsx, .xls, .csv)	Each sheet is cut into pages of **34 rows × 7 columns**.	Wide sheets continue on extra pages in reading order.
PowerPoint (.pptx)	Slides are rendered one by one as board pages.	Only the file’s index is read first; media over 3 MB is replaced with a placeholder.
PDF	Lazy byte-range pipeline described above.	Best choice for textbooks and scanned notes.

Support libraries (zip reader, Word and Excel converters) are fetched only when you first import that file type, so the app starts quickly and offline use of the board itself is unaffected.

25. Limits, Tips & Troubleshooting

Tips for the Smoothest Import

- Import one big file at a time and wait for pages to appear.
- Scroll steadily; pages wake as they come into view.
- Close other heavy browser tabs on tablets.
- Annotate first, export once: Export reads page after page.

If Something Looks Off

- A blank sleeping page fills in within a moment of scrolling.
- Very large images or media may be shown as placeholders.
- Use **Refresh** only if needed; it asks for confirmation.
- Re-import the file if a page fails to render.

26. Canvas Themes, Page Styles & Graph Papers

KryloCanvas supports 3 contrast themes, a custom theme colour, 4 standard page styles and 5 millimeter engineering graph colours.

Themes & Contrast Inversion

- **White Theme:** clean background (`#ffffff`).
- **Black Theme:** dark mode (`#000000`); black strokes invert to white automatically.
- **Grey Theme:** neutral studio background (`#4a4a4a`).
- **Custom Theme Colour:** any colour from the spectrum; grid, lines and dots adapt.

Paper Styles & 5 Graph Papers

- **Blank, Standard Grid, Single Lines** (ruled) and **Dotted** grid.
- **5 Millimeter Graph colours:** Green, Blue, Red, Orange and B&W.
- **Rotate, Delete and Select Page** keep multi-page boards tidy.
- **Export:** PDF (all / current page) or image, with optional rename.

27. Official System Access & Software Credentials

KryloCanvas is freely accessible worldwide as a lightweight web application. To launch the full whiteboard studio:

Direct Web Deployment

krylocanvas.vercel.app

Point your smartphone, tablet or interactive flat panel camera at the QR code on the left to immediately load the whiteboard studio. Compatible with Chrome, Safari, Edge, Firefox, Android and iOS.

Website URL: **https://krylocanvas.vercel.app**

This guide: **krylocanvas.vercel.app/KryloCanvas-Instruction-Guide.pdf**

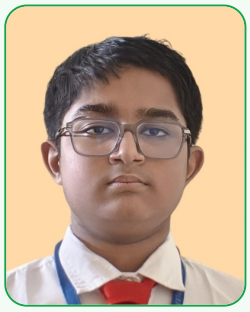
Software Architecture & System Engineering

About the Creator: Abhimat Chatterjee is an independent developer interested across multiple fields including **Robotics, STEM disciplines, algorithms, and interactive educational software**.

System Development: Driven to overcome the clunky latency and lack of precision in traditional digital boards, he conceptualized, designed, and coded from scratch as a lightweight, zero-dependency web application. It features a high-performance vector rendering pipeline, mathematical boundary-projection snapping for physical instruments, an extensible metric compass, dynamic flowchart trees, 43 geometric shapes and a lazy-loading engine for very large files.

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This instruction manual, system specification, and the KryloCanvas software engine are the proprietary intellectual property of the creator. No portion of this code, interface architecture, mathematical snapping algorithms, or documentation may be copied, distributed, sold, or replicated without explicit written permission from the creator.

Abhimat Chatterjee
LEAD ARCHITECT & CREATOR



Photograph of the Creator

KryloCanvas (Web-WhiteBoard) • Built to empower teachers, inspire students, and elevate classroom instruction through mathematical rigor and digital simplicity.